

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

**ENVIRONMENTAL SCIENCE****Paper 2 (Generated practice paper - Set 11)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

**Time allowed: 3 hours****Materials**

For this paper you may use:

- a calculator.

**Instructions**

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

**Information**

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| TOTAL    |      |

**Answer all questions in the spaces provided.**

**0 1**

The Isle of Eigg (area 31 km<sup>2</sup>, population about 100) is not connected to the national grid. Table 1 shows its electricity sources.

**Table 1**

| Source                       | Capacity / kW |
|------------------------------|---------------|
| Hydroelectric (3 generators) | 110           |
| Solar PV                     | 50            |
| Wind (4 turbines)            | 24            |
| Diesel generators (back-up)  | 160           |

**0 1**

**1**

Explain why Eigg uses a mix of three renewable resources rather than one, and why diesel back-up is still needed.

**[3 marks]**

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**0 1**

**2**

Describe **three** other ways in which the Eigg community has increased its sustainability.

**[3 marks]**

**1**

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**2**

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**3**

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**0 1**

**3**

Explain why small self-contained communities often find it easier to adopt sustainable lifestyles than large cities.

**[2 marks]**

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0 1 . 4

Explain how large-scale DC electricity grids such as Desertec could increase the use of renewable energy.

[2 marks]

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0 2

Table 1 shows some of the IUCN criteria used to place species in Red List categories.

Table 1

| Category              | Population decline over 10 years or 3 generations | Area of habitat            | Number of mature adults | Probability of extinction in the wild |
|-----------------------|---------------------------------------------------|----------------------------|-------------------------|---------------------------------------|
| Vulnerable            | over 40%                                          | under 2000 km <sup>2</sup> | under 10 000            | over 10% in 100 years                 |
| Endangered            | over 60%                                          | under 500 km <sup>2</sup>  | under 2500              | over 20% in 20 years or 5 generations |
| Critically endangered | over 80%                                          | under 10 km <sup>2</sup>   | under 250               | over 50% in 10 years or 3 generations |

0 2 . 1

State **three** roles of the IUCN.

[3 marks]

1 .....

2 .....

3 .....

0 2 . 2

A species of lemur has declined from 12 000 to 2 100 mature adults in three generations and occupies 400 km<sup>2</sup> of forest. Use Table 1 to determine its Red List category. Explain your answer.

[3 marks]

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0 2 . 3

Explain why only a small proportion of species have been assessed for the Red List, and give **two** criteria used to decide which species to assess.

[3 marks]

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0 2 . 4

Give **one** example of a species whose Red List category improved because of conservation and state the actions responsible.

[1 mark]

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10

03 . 1

Explain why tropical rainforests have such high biodiversity.

[4 marks]

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03 . 2

Explain why rainforest species may be particularly vulnerable to environmental change.

[2 marks]

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03 . 3

Explain why rainforest plants and animals such as frogs are a rich source of medicines.

[2 marks]

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03 . 4

Explain the importance of tropical rainforest for the hydrological cycle and for soils.

[2 marks]

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**0 4** Biomimetics uses knowledge of the adaptations of other species to improve the design of manufactured items.

**0 4** . **1** Complete Table 1 by adding the missing adaptation or application.

**[4 marks]**

**Table 1**

| Species and adaptation                               | Design application  |  |
|------------------------------------------------------|---------------------|--|
| Tubercles (bumps) on the flippers of humpback whales |                     |  |
|                                                      | Self-cleaning glass |  |
| Shark skin scales that reduce friction               |                     |  |
|                                                      | Velcro              |  |

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**0 4** . **2** Explain how the design of termite mounds has been copied to reduce the environmental impact of large buildings.

**[3 marks]**

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0 4 . 3

Describe **two** ways in which studying the physiology of other species could help to solve human health problems.

[2 marks]

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2 .....

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0 4 . 4

Explain why bird wing bones are a useful model for engineers.

[1 mark]

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0 5

Table 1 shows food conversion ratios (FCR) for livestock: the mass of feed needed to produce 1 kg of live mass gain.

Table 1

| Livestock      | FCR / kg feed per kg gain |
|----------------|---------------------------|
| Farmed salmon  | 1.2                       |
| Chicken        | 1.8                       |
| Pork           | 3.0                       |
| Beef (feedlot) | 7.0                       |
| Sheep          | 8.0                       |

0 5 . 1

Use Table 1 to explain why intensive poultry and fish production convert feed to food more efficiently than beef.

[3 marks]

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**0 5 . 2**

Explain **three** ways farmers reduce food chain energy losses in livestock production.

**[3 marks]**

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**0 5 . 3**

Explain why growing crops for direct human consumption usually produces more food per hectare than rearing livestock, and describe **one** situation where livestock is the better option.

**[4 marks]**

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**10**

06 . 1

Give **two** natural processes that are driven by solar energy.

[2 marks]

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06 . 2

Explain why natural processes can take place at low temperatures.

[2 marks]

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06 . 3

Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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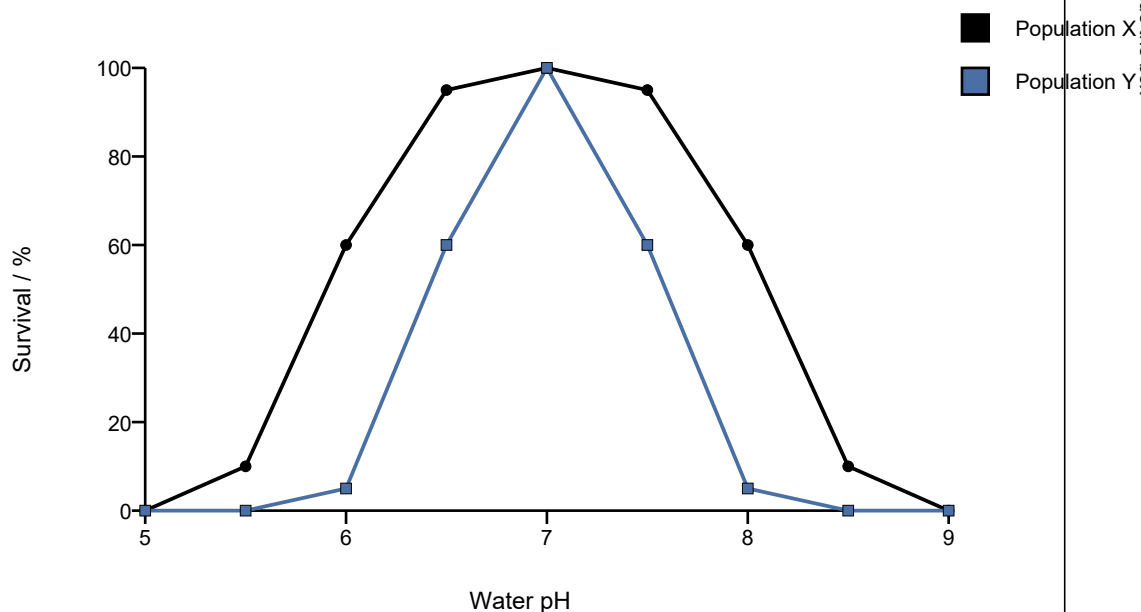
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07

Figure 1 shows the percentage survival of two populations of a freshwater snail species (X and Y) at different water pH values.

**Figure 1**



07

1 Explain what is meant by the term **range of tolerance** and use Figure 1 to compare the ranges of tolerance of populations X and Y.

**[3 marks]**

07

2 A stream is acidified by mine drainage from pH 7.0 to 6.0. Use Figure 1 to explain which population is more likely to survive and why.

**[3 marks]**

07 . 3

Explain how an understanding of a species' range of tolerance helps in conservation management.

[2 marks]

07 . 4

Give **two** biotic factors that affect the distribution of a species.

[2 marks]

1

2

10

08 . 1

Describe coppicing and explain how it increases the biodiversity of a woodland.

[4 marks]

08 . 2

Explain how the reintroduction of grey wolves to Yellowstone National Park increased biodiversity.

[4 marks]

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**08 . 3**

Explain why culling or removing species may be necessary in habitat management.

**[2 marks]**

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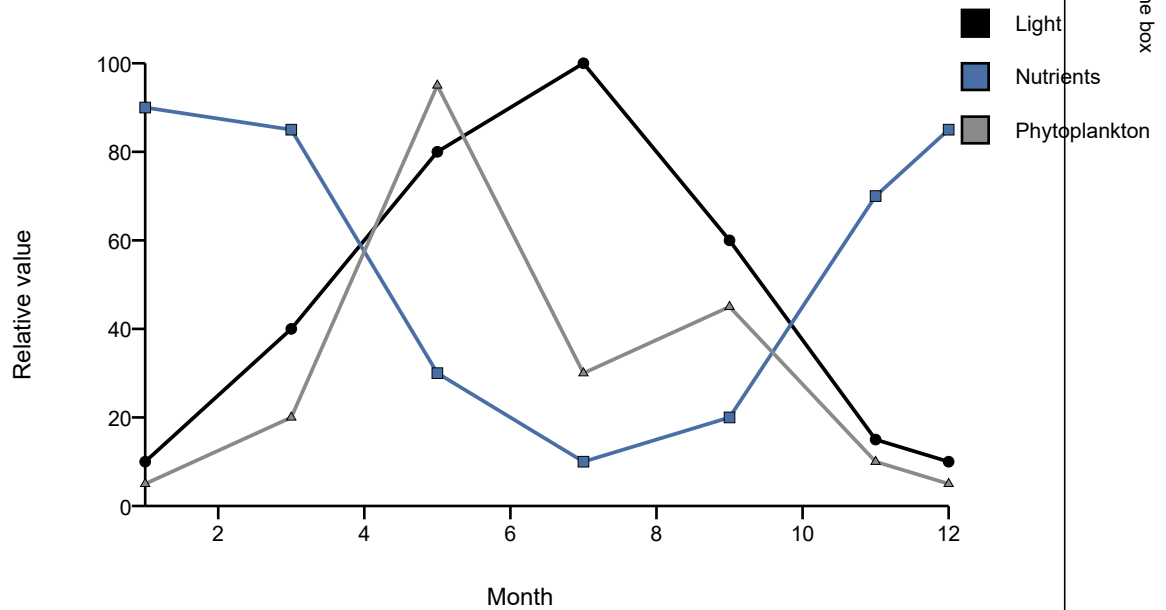
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**10**

09

Figure 1 shows the seasonal changes in light intensity, nutrient concentration and phytoplankton biomass in the surface water of a temperate ocean in the northern hemisphere.

Figure 1



09

1 Use Figure 1 to explain the spring peak and summer decline in phytoplankton biomass.

[4 marks]

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09

2 Explain why tropical open oceans have low productivity all year despite high light and temperature.

[2 marks]

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09 . 3

Explain why areas of upwelling and coastal waters are the most productive parts of the sea.

[3 marks]

09 . 4

Explain why phosphate is often the limiting nutrient in the sea.

[1 mark]

10

10 . 1

Discuss the strategies that can be used to increase the sustainability of agriculture.

[9 marks]

[illegible]

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**10 . 2** State what is meant by **permaculture**.

**[1 mark]**

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**10**

**Write an essay on one of the following topics.**

$$\boxed{1} \boxed{1} . \boxed{1}$$

Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

**[25 marks]**

**OR**

$$\boxed{1} \boxed{1} . \boxed{2}$$

Discuss how human activities influence biodiversity, using examples from a range of different contexts.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.





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**END OF QUESTIONS**